

**10Pearls**

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# **Full-Stack Assessment**

AI-Powered Consultation Workflow

Expected Completion: **2 Days**

*Innovate, modernize, accelerate.*

# Objective

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Build a simplified version of an AI-powered consultation workflow that allows clinic administrators to:

- Manage consultation records
- Interact with an AI assistant during consultations
- Generate treatment recommendations
- Book appointments and update system records

This assessment evaluates your ability to implement a complete end-to-end product flow, not just isolated screens.

# Expected User Flow (IMPORTANT)

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Your system must support the following journey:

|   |                                               |
|---|-----------------------------------------------|
| 1 | Dashboard                                     |
| 2 | Consultation Records (List)                   |
| 3 | Consultation Detail (AI Chat – In Progress)   |
| 4 | Consultation Summary (AI Output)              |
| 5 | Appointment Booking                           |
| 6 | Back to Consultation Records (Updated Status) |

# Functional Requirements

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## 1. Dashboard (Overview)

Display high-level metrics:

- Total consultations
- Booked appointments
- Conversion rate
- Navigation sidebar with "Consultations" entry point

## 2. Consultation Records

Table of consultations with:

- Patient name
- Primary concern
- Recommended procedure
- Status

Support:

- Filtering (Pending, Booked, Completed)
- Search
- Clicking a record opens consultation details

## 3. Consultation Detail (AI Chat)

Simulate an AI-assisted consultation:

- User enters queries
- System returns responses (mock or real)

Must support:

- Conversation state persistence
- Multi-message interaction

AI responses may include:

- Structured data (tables, insights)

#### 4. Consultation Summary

Generated after consultation completes. Display:

- Patient summary (from conversation)
- Recommended treatments
- AI reasoning (optional)

Provide:

- "Restart Consultation"
- "Book Appointment"

#### 5. Appointment Booking

Form to capture:

- Selected treatment
- Date and time
- Location

On confirmation:

- Save appointment
- Update consultation status → "Booked"
- Redirect to Consultation Records

### Technical Requirements

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You can use any frontend and backend stack that you are trained on.

#### Data & Persistence

Store:

- Consultation sessions
- Messages
- Recommendations
- Appointments

Maintain linkage between:

- consultation → recommendation → booking

### Evaluation Criteria

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| Area                           | Weight |
|--------------------------------|--------|
| End-to-End Flow Implementation | 25%    |
| Backend Design & APIs          | 20%    |
| Frontend Architecture & UX     | 15%    |

|                              |            |
|------------------------------|------------|
| State Management Across Flow | <b>15%</b> |
| Data Modeling & Persistence  | <b>15%</b> |
| Code Quality                 | <b>10%</b> |

## Bonus (Optional)

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- AI integration (OpenAI / Azure OpenAI)
- Dynamic conversation branching
- Clean architecture separation (AI service vs API)
- Deployment (Azure / Docker)

## Submission

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- GitHub repository

README must include:

- Setup instructions
- Architecture overview
- Assumptions
- Demo video or presentation to the Architect on the project

## Timeline

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Expected completion: 2 Days

## Additional Notes and Expectations

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### 1. API Design Expectations

Candidates are expected to design and implement well-defined REST APIs for the system, including (at minimum):

- Consultation lifecycle management
- Chat/message handling
- Recommendation generation
- Appointment booking

APIs should be:

- Clearly structured and logically grouped
- Consistent in request/response design
- Suitable for scaling in a real-world system

### 2. Data Modeling Expectations

Candidates must define and implement an appropriate data model covering:

- Consultations (sessions)
- Messages (chat history)
- Recommendations
- Appointments

Data relationships must be maintained (e.g., a consultation should link to its messages, recommendations, and booking).

### **3. State Management (Critical Requirement)**

The system must demonstrate stateful behaviour across the consultation flow. Specifically:

- Consultation progress should persist across multiple interactions
- Users should be able to move between screens without losing context

The system should maintain continuity from:

- Chat → Summary → Booking → Records

### **4. AI Interaction Expectations**

AI integration may be mocked or real, but it must demonstrate:

- Context-aware responses (based on prior messages)

At least one structured output, such as:

- A data table
- A list of recommendations

A transition from:

- Unstructured input → structured outcome (summary/recommendations)

### **5. Error Handling & Resilience**

The system should handle common failure scenarios gracefully, including:

- API failures
- Invalid or missing inputs (e.g., booking form)
- Interrupted flows (e.g., incomplete consultation)

Basic user-friendly feedback must be provided.

### **6. What to Avoid**

To ensure meaningful evaluation, avoid the following:

- Hardcoded data in place of real APIs
- Single-file or tightly coupled backend implementations
- UI-only solutions without backend persistence
- Ignoring flow transitions between screens

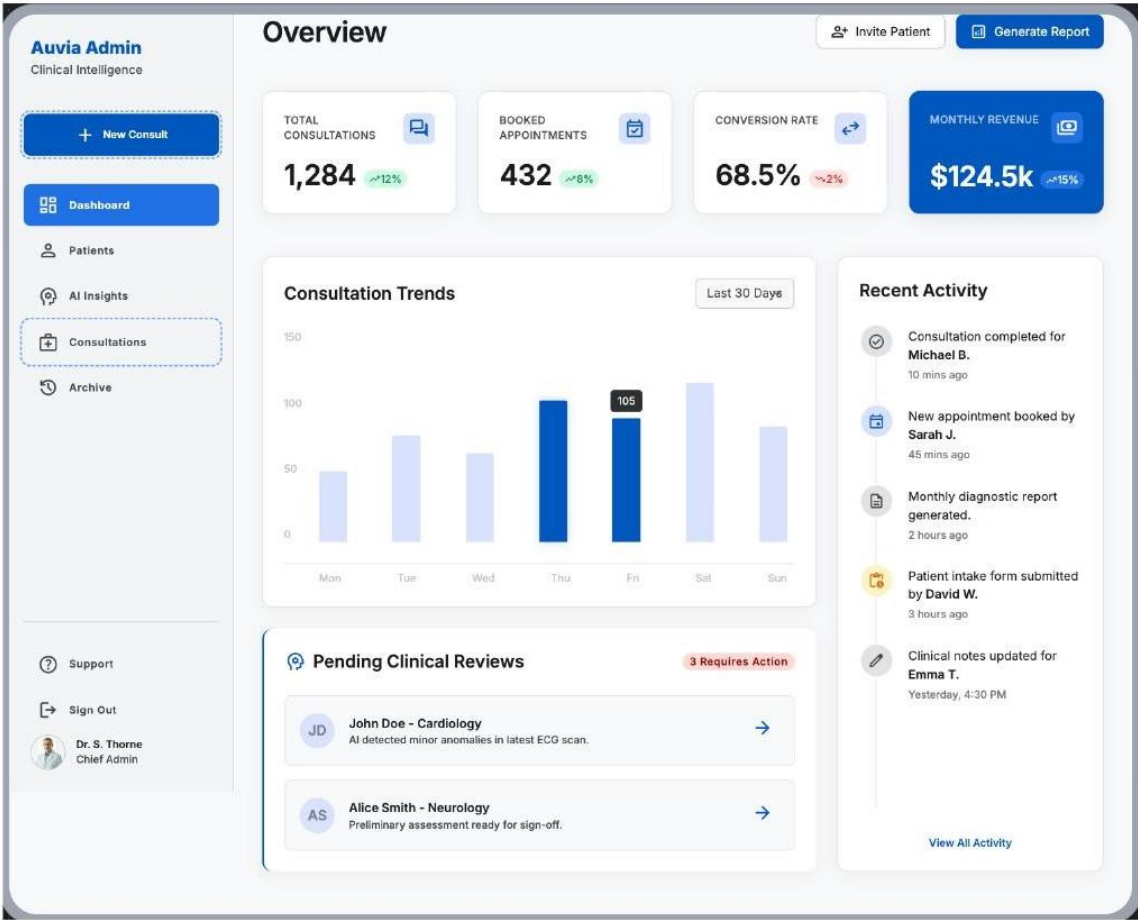
# Application Screens (Web)

Total 5 screens. The dotted light blue lines around tabs indicate the logical flow to the next screen and back.

## Screen 1 — Dashboard (Overview)



Screen 1



## Screen 2 — Consultation Detail (AI Chat)



### Screen 2

**Auvia Admin**  
Clinical Intelligence

+ New Consult

Dashboard

Patients

AI Insights

**Consultations**

Archive

Support

Sign Out

Sarah Jenkins  
AI Consultation in Progress

Can you provide the specific troponin trend over the last three visits?

DR

Certainly. Here is the requested trend data. Note the sharp incline in the most recent draw.

| Date         | Value (ng/mL) | Status   |
|--------------|---------------|----------|
| Oct 12, 2023 | 0.01          | Normal   |
| Jan 05, 2024 | 0.03          | Normal   |
| Today        | 0.09          | Elevated |

What are the risks?

Draft referral to Cardiology

Compare with previous ECG

...

Ask Auvia about Sarah's data...

Auvia AI can make mistakes. Verify clinical information.

Screen 3 — Consultation Summary (AI Output)



Screen 3

Auvia Admin

Clinical Intelligence

New Consult

Dashboard

Patients

AI Insights

Consultations

Archive

Support

Sign Out

Patient Summary

Primary Concern: Fine Lines

Patient reported increasing visibility of fine lines around the periorbital and nasolabial areas over the past 12 months.

Secondary Concern: Skin Texture

Desires improvement in overall skin surface texture and mild uneven pigmentation.

Medical History Note

No contraindications reported. Previous clear tolerance to topical retinoids.

AI INSIGHT

Algorithm indicates high probability of patient satisfaction combining structural volume restoration with superficial resurfacing, given the specific combination of fine lines and texture concerns.

Confidence: 94%

Recommended Treatments

Dermal Fillers

Priority 1

Targeted hyaluronic acid injections to address nasolabial folds and restore mid-face volume, directly addressing the primary concern of fine lines and structural support.

Est. Range\$600 - \$1,200

Laser Resurfacing

Priority 2

Fractional non-ablative laser therapy to improve overall skin texture, stimulate collagen production, and address mild uneven pigmentation over 3-4 sessions.

Est. Range (Per Session)\$400 - \$800

Next Steps

Review these recommendations with the patient. You can adjust the treatment plan during the final consultation phase.

Book Appointment

Restart Consultation



Screen 4 — Appointment Booking



Screen 4

Auvia Admin

Clinical Intelligence

Dashboard

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New Consult

Support

Sign Out

Back to Patient Record

### 3. Appointment Booking

Configure appointment details for the recommended clinical procedure.

Procedure Details

Primary Treatment

Dermal Fillers

Target Area

Nasolabial Folds

Estimated Duration

45 Minutes

AI Recommendation

Based on patient history, recommend scheduling 2 weeks post-consultation to allow for pre-treatment protocols.

Logistics

Clinic Location

Downtown Medical Center (Primary)

Date

dd/mm/yyyy

Time

--:-- --

Provider (Optional)

No Preference - First Available

Summary

Patient

Emma Richardson

Procedure

Dermal Fillers

Location

Downtown Medical Center

Estimated Cost

\$850.00

Final price may vary based on exact units used.

Confirm Appointment

Save as Draft

Screen 5 — Consultation Records



Screen 5

**Auvia Admin**  
CLINICAL INTELLIGENCE

Dashboard

Patients

AI Insights

Consultations

Archive

+ New Consult

Support

Sign Out

### 4. Consultation Records

Review and manage historical patient consultations and AI-assisted procedure recommendations.

Search patient or ID...

FILTER BY STATUS:

AllPendingBookedCompleted

Sort by Date (Newest)

| PATIENT NAME                | PRIMARY CONCERN                   | RECOMMENDED PROCEDURE                  | STATUS    | DATE   |
|-----------------------------|-----------------------------------|----------------------------------------|-----------|--------|
| <div>SJ</div> Sarah Jenkins | Atypical chest pain, palpitati... | <div>24-Hour Holter Monitor</div>      | Pending   | Oct 24 |
| <div>MC</div> Marcus Chen   | Chronic cluster migraines         | <div>Neurology Referral (MRI)</div>    | Booked    | Oct 22 |
| <div>ET</div> Elena Torres  | Post-operative joint stiffness    | <div>Physical Therapy Assessment</div> | Completed | Oct 18 |
| <div>DR</div> David Ross    | Routine metabolic screening       | <div>Comprehensive Blood Panel</div>   | Completed | Oct 15 |

Showing 1 to 4 of 48 records